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ChE 321

Homework 8

P12_16B

A)

$A \rightleftharpoons B$

$v_0 = 1\ dm^3/min$

$F_{A0} = 10\ mol/min$

$V = 10\ dm^3$

$\tau = \frac{V}{v_0} = \frac{10}{1} = 10\ min$

$UA = 3600\ cal/min$

$T_a = T_0 = 37^\circ C = 310\ K$

$C_{p,A} = C_{p,B} = 40\ cal/mol \cdot K$

$C_{p_0} = 40\ cal/mol \cdot K$

$k(400K) = 1 min^{-1}$

$k(400K) = 1\, min$

$K_c(400K) = 100$

$E/R = 20{,}000\, K$

$F_{A0} - F_A - r_A V = 0$

$$X = \frac{F_{A0} - F_A}{F_{A0}}$$

$\Rightarrow F_A = F_{A0}(1 - X)$

$A \rightarrow B \; (1:1)$

$C_A = C_{A0}(1 - X)$

$$C_B = C_{A0} X$$

$$r_A = k_f C_A - k_r C_B$$

$$r_A = k C_A - \frac{k}{K_c} C_B$$

$$= k C_{A0} \left[(1-X) - \frac{X}{K_c} \right]$$

$$F_{A0} X = r_A V$$

$$\boxed{X(T) = \frac{k(T)\tau}{1 + k(T)\tau\left(1 + \frac{1}{K_c(T)}\right)}}$$

$$\frac{dT}{dt} = G(T) - R(T)$$

$@T = 400K$

$$X(400) = \frac{10}{1+10(1.01)}$$

$$= \frac{10}{1+10.1} = \frac{10}{11.1} \approx 0.90$$

$$r_A = \frac{10(0.90)}{10} = 0.90$$

$$G(400) = \frac{80000(0.9)(10)}{10(40)}$$

$$= \frac{720000}{400} = 1800\ K/min$$

B)

```
   T_ss_K        X_ss
   ______     __________

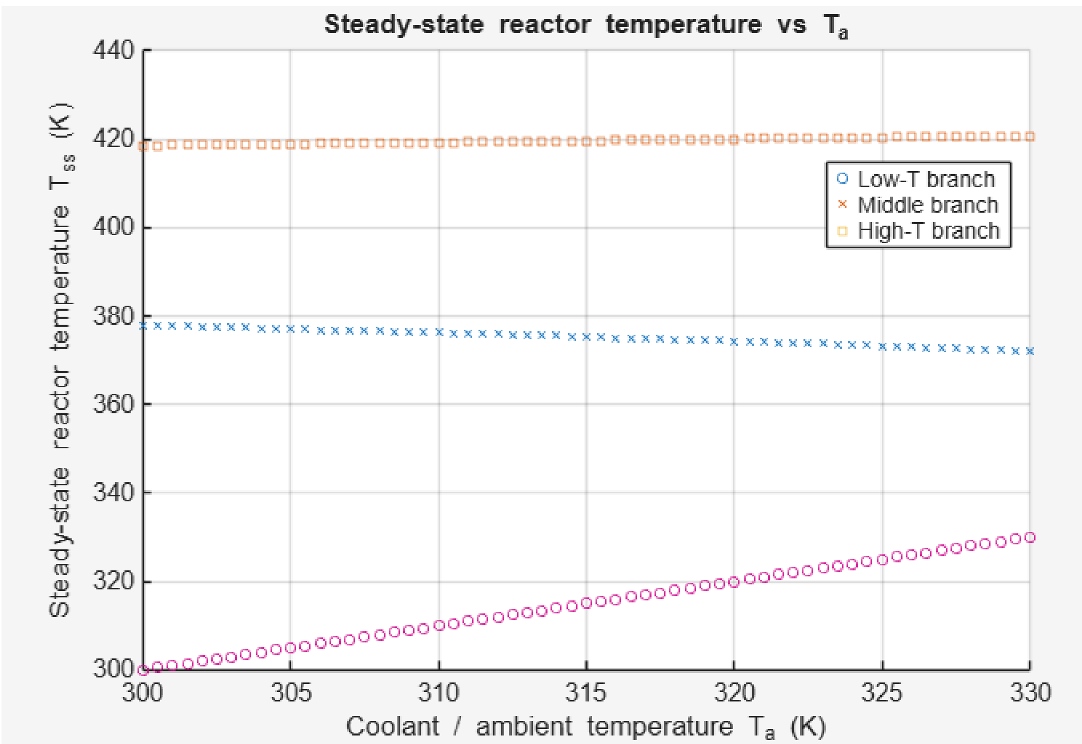
      310     4.9639e-06
   376.23        0.29802
   419.28        0.49176
```

C)

```
T_ss = 310.001 K:  d(G-R)/dT = -8.998e+00  -->  STABLE
T_ss = 376.228 K:  d(G-R)/dT = 5.012e+01  -->  UNSTABLE
T_ss = 419.280 K:  d(G-R)/dT = -1.218e+02  -->  STABLE
```

D) X = 0.49176

E)




```
Ignition T_a ≈ 300.00 K
Extinction T_a ≈ 330.00 K
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F)

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Adiabatic new upper steady state
  T_new_upper   = 431.48 K
  X_new_upper   = 0.0607
```

G)

```
UA that maximizes conversion on upper steady state
  UA_opt   = 7800.00 cal/(min·K)
  T_opt    = 404.27 K
  X_max    = 0.9192
```

H) We want to increase conversion beyond the current upper steady state by changing the coolant temperature only and the heat exchanger product. Based on the T_{ss} vs T_a and X vs UA behavior for the reactor, propose a strategy to increase the conversion while maintaining safe and stable operating conditions.

This requires critical thinking because there is no single formula that gives the answer.

I)

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Adiabatic blowout flow rate
  v0_blowout = 2000.000 dm^3/min
  T_blowout  = 310.00 K
  X_blowout  = 0.0000
```

J) Use a relatively large UA so that heat removal is strong.